

NOETHER: A Constructive Framework for Metamorphic Pattern Discovery from Operator Algebras

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Metamorphic Testing (MT) is recognised in IEEE/ISO software-testing standards and is widely recommended for testing AI systems. Its progress, however, remains constrained by metamorphic relation (MR) identification, a step that depends on tester domain knowledge and yields MR sets difficult to reuse across teams and domains. Existing approaches, including structured frameworks (METRIC, METRIC+), mining and evolutionary pipelines (MR-Scout, GenMorph), LLM-assisted methods, and MetaPattern catalogues, share an inductive grounding that leaves three foundational questions open: *origin*, *closure*, and *transferability*.

This paper introduces **NOETHER**, a two-layer framework for MetaPattern discovery. The *downstream layer* is mechanical and provable: given an operator-algebra block decomposition, the CONSTRUCT-MP algorithm produces a MetaPattern set with an algebraic-closure guarantee under the framework's Translate operator (Theorem 1) and polynomial-time decidability under a finite generating set (Theorem 2); absolute completeness over arbitrary properties expressible in the operator algebra is identified as an open conjecture. The *upstream layer* is a curated eight-block decomposition over recurrent mathematical structures (symmetry, order, self-adjoint, time-reversal, limit, qualitative-dynamics, method-comparison, and relational equivalence), stated as an explicit empirical hypothesis with a documented out-of-scope catalogue rather than as an axiom.

The framework's scope precondition is that the program family under test admits an explicit operator-algebraic description through mathematical or physical equations; programs lacking such structure are out of scope by construction. Within this scope, we instantiate NOETHER on three structurally distinct domains: a Boltzmann reactor-physics transport solver, where the framework systematises a prior inductive catalogue and re-classifies further equivalence classes; equivariant machine learning, where executable MRs are derived for rotation invariance, adjoint duality, and training-trajectory reversibility; and relational query optimisers, whose idempotent-semiring algebra exercises the relational-equivalence block. Structural-coverage predictions are consistent with the data within each case-study scope; comparative evaluation against three SOTA representatives (a GP-evolved baseline, an LLM-assisted baseline, and a mining-based baseline) is reported alongside an algebra-aligned metric framework that distinguishes algebra-disrupting from algebra-preserving mutants and an MR-generation cost matrix; a head-to-head superiority claim is not asserted. A negative instantiation on a fourth program family (PWR core simulators) falsifies the absolute-completeness conjecture (Theorem 1') on a structurally significant operator algebra, via two independent counterexamples that jointly identify five pairwise-independent extensions of the framework's Translate operator as the principal locus of follow-up work. NOETHER lifts induction from per-program MR sampling to a stable per-domain algebraic layer and makes the downstream step deductive and mechanical without eliminating induction at the upstream layer.

CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation*; • **Theory of computation** → *Algebraic semantics*.

Additional Key Words and Phrases: metamorphic testing, metamorphic relation identification, MetaPattern, operator algebra, equivariance, algebraic completeness, software testing foundations

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1 Introduction

Noether's first theorem replaced an empirically curated catalogue of conservation laws with a derivation from the structure of an action functional.¹ A catalogue of observed invariants can sometimes be replaced by a derivation procedure grounded in the underlying structure. The replacement does not eliminate empirical work; it lifts the empirical step from per-instance enumeration to per-domain structural identification, one cycle of induction per stable domain rather than per program.

Software testing faces a related problem. Metamorphic Testing (MT), introduced by Chen et al. in 1998 [10], was admitted into the IEEE/ISO/IEC software-testing standards in 2022 [22] and has been increasingly endorsed as a technique for testing AI and machine-learning systems [27, 39]. Its central artefact is the *metamorphic relation* (MR): a property that constrains how a program's outputs must covary across multiple executions, thereby substituting for an explicit oracle in domains where one cannot exist. Despite MT's maturity, MR identification, the task of deciding *which* properties hold for a program under test, remains its binding constraint. Practitioners report three recurring difficulties: high dependence on tester domain knowledge, mutually incompatible MR formulations for the same program, and low reuse of MR sets across teams or projects [27, 39]. Recent work has responded at the application layer, by mining MRs for particular domains, and at the integration layer, by automating search through evolutionary, mining-based, or LLM-assisted pipelines [4, 41, 53]. The foundational layer has not advanced at the same pace.

The principal artefact at that foundational layer is the *MetaPattern* (MP): an equivalence class over MRs that captures a recurrent structural strategy a tester invokes when reasoning about program properties. MetaPatterns organise the otherwise unbounded MR design space into a small, interpretable scaffold; methods that exploit them, including the structured MR identification approaches METRIC [11] and METRIC+ [45], and the recent wave of LLM-prompted MR generators, depend on the scaffold's quality. Yet across the literature, MetaPattern catalogues continue to be assembled in the same way conservation laws were assembled before 1918: by induction over observed examples. A typical proposal lists k patterns drawn from cluster analysis or expert codification, demonstrates that the patterns "cover" some corpus of MRs to a target threshold, and stops. None of the existing MP proposals, including the authors' own prior work on five reactor-physics patterns, answers the three questions that any foundational theory of MetaPatterns should answer:

- (1) **Origin.** *Why* exactly these MetaPatterns and not others? What is the structural source of an MP, as distinct from an empirical regularity in the corpus on which it was induced?
- (2) **Closure.** Under what mathematical conditions is a discovered MP set *closed under a stated derivation operator*, in the sense that it is guaranteed not to miss patterns reachable through the operator from a structurally fixed input? Absolute completeness over all properties one might write over the underlying structure is a strictly stronger demand and remains open in general.
- (3) **Transferability.** When the program family changes (from reactor physics to natural-language inference, from numerical libraries to recommender systems), how does the MP set change, and can the new set be obtained without re-running the entire empirical induction in the new domain?

We call this the *origin–closure–transferability gap*. The gap matters because it explains why MR sets continue to grow as one-off artefacts. Without a structural source, there is no clear boundary

¹The theorem appeared in Noether's 1918 paper *Invariante Variationsprobleme*, addressing a question Hilbert and Klein had raised about energy conservation in general relativity. We do not invoke it as a theorem about programs: program semantics and metamorphic relations do not provide an action functional. The analogy is methodological only.

on what the pattern space contains. Without a transfer rule, a relation written for one program rarely moves cleanly to another, even when both programs share a deep structural template. These are not only tooling limitations; they follow from grounding MetaPatterns in observed examples rather than in the structure that makes the relations hold.

This paper proposes such a structural source. We introduce **NOETHER**, a constructive framework that derives MetaPatterns from the operator-algebraic structure of the program family under test. The framework extracts MetaPatterns from invariants of an operator algebra $\mathcal{A}_P$ (formally defined in Section 3) that captures the program family’s mathematical scaffolding. Given $\mathcal{A}_P$, NOETHER produces a MetaPattern set $M(\mathcal{A}_P)$ (constructed by the algorithm of Section 4) together with a closure guarantee over the algebra-induced MR space. Given a different program family with a different algebra $\mathcal{A}_{P'}$, the same construction produces $M(\mathcal{A}_{P'})$ without re-running empirical induction. The framework does not abolish induction: domain experts must still distil $\mathcal{A}_P$ from program semantics, and Section 8 states this limitation explicitly. What becomes algebraic is the downstream step from $\mathcal{A}_P$ to the MetaPattern set.

We make four contributions.

- **C1.** We introduce NOETHER, a *two-layer* framework for MetaPattern discovery that combines an empirically curated eight-block decomposition (Hypothesis 1; upstream layer) with a constructive algorithm for deriving MetaPatterns from a program-induced operator algebra (downstream layer).
- **C2.** We prove an Algebraic Closure Theorem (Theorem 1) for the constructed set: given the eight-block decomposition, the resulting MetaPattern set is closed over the algebra-induced MR space under the framework’s Translate operator. We also establish polynomial-time decidability when the algebra admits a finite generating set (Theorem 2). We further establish that the strictly stronger statement of absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$ (Theorem 1’ / Conjecture D) is false on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$: two specific MRs from the standard PWR safety-analysis literature (non-additivity of rod-bank reactivity worth, and second-order mixed dependence of k_{eff} on moderator temperature and boron concentration) are formulable over $\mathcal{A}_{\text{PWR}}$ but not in $\text{MR}(\mathcal{A}_{\text{PWR}})$ (§6.8, Appendix D). The two MRs identify five pairwise-independent structural obstructions in Translate’s signature (operator-spectrum output, homomorphism-failure π -template, configuration-indexed adjoint structure, higher-order mixed-difference templates, two-direction joint parametric dependence) as the principal open problem for follow-up work. The scope of each result is stated explicitly in Section 4.3.
- **C3.** We instantiate NOETHER on a real-world program family and show how it *systematises and re-classifies* previously catalogued patterns. The framework reproduces three prior MetaPatterns, refines two on a sounder algebraic basis, and includes a deflationary correction direction (revealing that the inductive grid was *over-counted* on some sub-families and *under-counted* by the absence of m_{adj} and m_{rev}). The two structurally distinct re-classified equivalence classes are not de novo discoveries; domain experts could have written them down. NOETHER’s contribution is the algebraic warrant for treating them as separate MetaPatterns within a uniform structure. The deflationary direction is non-circular relative to the prediction caveat of §5.3: §8.3 demonstrates it on three independent test cases (a sorting library; Murphy et al. 2008’s six-class ML categorisation [32] re-decoded against a feedforward classifier algebra; the prior reactor-physics catalogue itself), without invoking T^* or $\mathcal{T}^*$ structure that was curated from reactor physics.
- **C4.** We demonstrate cross-domain transferability *within the framework’s scope precondition* (programs admitting an explicit operator-algebraic description) by instantiating NOETHER

on three structurally distinct domains: Boltzmann reactor-physics transport, equivariant machine learning (Section 6), and relational query optimisers (Section 6.7), the last exercising the relational-equivalence block whose algebraic skeleton differs from the Lie-group / self-adjoint / time-reversal core. On a small-scale comparative case study and a DeepCrime-style real-fault pilot ($n = 5$, underpowered for $\alpha = 0.05$ inferential conclusions), the framework's structural-coverage prediction is consistent with the data within case-study scope; the case study's category-(iv) detection contrast exhibits construct validity of $\rho_{\text{train-rev}}$ as a gradient-reversal probe rather than averaged superiority, since the mutation set was constructed to cover one defect category per non-empty block of $\mathcal{A}_{\text{equi}}$. A larger comparative evaluation against three SOTA representatives (one per category: GP-evolved, LLM-assisted, mining-based) is reported as a pre-registered protocol.

Scope of contribution. This is a theoretical paper, and its contribution is systematisation rather than deduction from first principles. The eight blocks are curated by inspecting mathematical structures that recur across the program families we have studied; they are not derived from an algebraic axiom. The framework therefore has two layers: an *upstream layer* (curating $\mathcal{A}_P$ and its block decomposition) that remains empirical and human, and a *downstream layer* (mechanically deriving $\mathbb{M}(\mathcal{A}_P)$ from $\mathcal{A}_P$) that is algorithmic and provable. We do not claim to have eliminated induction from MetaPattern discovery. We move induction one level up, from “what MetaPatterns recur in observed MR samples?” to “what algebraic structures recur in the program families practitioners care about?”, and make the downstream step mechanical. The engineering payoff of this re-grounding awaits empirical follow-up work; comparative evaluation against existing automated MR-identification pipelines is reported as a protocol in Section 6.6 and constitutes part of the framework's resubmission obligations.

Boundary of contribution

This paper establishes:

- (1) Algebraic closure of $\mathbb{M}(\mathcal{A}_P)$ under Translate for the operator algebras stated, given a block decomposition (Theorem 1);
- (2) Polynomial-time decidability of CONSTRUCT-MP under explicit complexity assumptions (Theorem 2);
- (3) Three non-vacuous instantiations: Boltzmann reactor physics (§5), equivariant ML (§6), and relational query optimisers (§6.7); the relational-equivalence block $\mathcal{B}_{\text{rel}}^*$ extends the framework beyond the Lie-group / self-adjoint / time-reversal mathematical core.

It does *not* establish:

- (a) Absolute completeness over arbitrary properties expressible in $\mathcal{A}_P$. This is Theorem 1' (Conjecture D, Appendix D); §6.8 establishes that it is false on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$ via two independent counterexamples (non-additivity of rod-bank reactivity worth, and second-order mixed dependence of k_{eff} on moderator temperature and boron concentration), but the question of whether a Composite-Translate extension absorbs the five obstructions identified by the counterexamples while preserving Theorem 1's closure and Theorem 2's polynomial-time decidability remains open.
- (b) Sufficiency of the eight-block list. Hypothesis 1 is an empirical hypothesis with six enumerated out-of-scope program-family classes (Remark 2); each such class is a candidate ninth block.
- (c) Superiority over existing automated MR-identification pipelines on average. The comparative evaluation in §6.6 establishes effects for specific defect categories on small benchmarks; it does not characterise the framework's behaviour on arbitrary defect distributions.
- (d) Elimination of induction from MetaPattern discovery. Induction is *relocated* from MR-instance level to algebra-block level, not eliminated.

The remainder of the paper is organised as follows. Section 2 surveys the four lines of prior work; Section 3 introduces the operator-algebraic preliminaries; Section 4 presents NOETHER itself, the construction algorithm, and the two theorems; Section 5 instantiates NOETHER on the Boltzmann transport equation; Section 6 cross-instantiates on equivariant ML; Section 7 reports a head-to-head empirical comparison against an automated SOTA baseline (GenMorph) on programs with explicit mathematical-physics equations; Section 8 discusses threats and relationships; Section 9 concludes. Appendix A instantiates NOETHER on the remaining reactor equations; Appendix B gives the source-equation, Translate-template and block-assignment provenance of all 12 representative MRs of Table 3; Appendix C gives worked multi-block-derivable MR examples; Appendix D contains the full proofs; Appendix E gives a Python reference implementation.

2 Background and related work

We organise prior work along four lines: MT/MR fundamentals and the long-standing identification bottleneck; structured MR identification through METRIC and METRIC+; automated MR identification, including MR-Scout, GenMorph, and LLM-assisted approaches; and MetaPattern catalogues, including our prior reactor-physics taxonomy. These lines differ in mechanism, but they share one limitation: their pattern structures are induced from observed MRs rather than derived from the mathematical structure that makes those MRs hold.

2.1 Metamorphic testing and the MR identification bottleneck

Metamorphic Testing was introduced by Chen, Cheung, and Yiu in 1998 to address the test-oracle problem [10]. An MR is a logical implication of the form $R_i(x_1, \dots, x_n) \Rightarrow R_o(P(x_1), \dots, P(x_n))$, where P is the program under test, R_i is an input relation, and R_o is an output relation. When R_i holds but R_o fails on actual executions, a fault is reported, without any need to know the absolute correct output of any single execution. Over two decades MT has become standard equipment for testing systems whose oracles are otherwise inaccessible: scientific computing, machine learning classifiers, autonomous vehicles, search engines, compilers, and large language models [27, 39].

The community has long acknowledged a single binding constraint on MT’s effectiveness: MR identification. Surveys spanning twenty years agree that identifying high-quality MRs requires (i) deep familiarity with the program’s functional semantics, (ii) substantive domain background (physical, mathematical, or linguistic, depending on the system), and (iii) the ability to convert that background into executable property assertions [27, 39]. Liu et al. [29] provide the canonical empirical evidence that even ad-hoc MRs detect a substantial fraction of mutants given modest tester training, while Murphy et al. [32] catalogued the structural properties of ML applications relevant to MR design and Xie et al. [52] demonstrated that MT scales effectively to supervised classifiers. Testers without sufficient domain background tend to identify only “trivial MRs”. A 2024 survey explicitly identifies AI assistance, especially via large language models, as the most promising open avenue for closing the identification gap [27]; Saha and Kanewala [37] report that, on a 709-mutant benchmark for supervised classifiers, current MR sets detect only 14.8%, underscoring the gap.

Beyond difficulty, the MR identification bottleneck has a more troubling structural feature: even when MRs are identified, the field has accumulated little consensus on *how* they are identified. Different authors confront the same program and produce dissimilar MR sets. Once written, an MR rarely migrates: a relation drafted for one ML classifier seldom transports to another, and a relation drafted for one numerical solver seldom transports across solver families.

2.2 Structured MR identification: METRIC and METRIC+

Structured MR identification has already moved the field away from ad hoc relation design. METRIC organises MR construction around an “input/output category” framework: the tester first identifies relevant categories of input transformations and output relations, then composes MRs from category pairs [11]. METRIC+ extends this scheme by enriching the category catalogue and providing systematic combination rules that reduce the human burden in category enumeration [45]. Both approaches are widely cited and remain the strongest existing attempt to give MR identification an explicit scaffold.

Our disagreement is not with this direction but with its grounding. The categories are introduced through expert curation and validated by empirical coverage on benchmark programs. This leaves two of our three foundational questions unanswered. *Origin*: METRIC and METRIC+ do not derive their categories from program-level mathematical structure. *Closure*: neither framework provides a mathematical condition under which the category set is guaranteed to be complete; coverage is reported but not proved. The third question, *transferability*, is only partially addressed. The same category templates can be invoked across domains, but because the categories are not algebraically bound to specific program structures, transfer rests on an assumption of universality that remains open.

2.3 Automated MR identification

Automated MR identification attacks the bottleneck by searching for executable relations more directly. MR-Scout mines MRs from existing test suites by extracting input-transformation and output-assertion patterns from test-case pairs and abstracting them into reusable relations [53]; closely related is the work of Nair et al. on mining MRs from bug-report corpora [33]. GenMorph evolves MR candidates through genetic programming, co-evolving input transformations with output assertions and using mutation-killing as the fitness signal [4]. For deep-learning pipelines specifically, MT4DL extends MR mining to the multi-stage training-and-inference pipeline [58]. Shin et al. derive executable MRs from natural-language requirements via few-shot prompting of a large language model, with validation through an industrial questionnaire study with Siemens [41]; the systematic empirical study of Xu et al. [54] characterises the practical limits of LLM-generated MRs across multiple ML benchmarks. Further LLM-assisted variants, including domain-customised GPTs for autonomous-driving simulators [57] and multi-agent retrieval-augmented pipelines for traffic-rule MRs [28], extend this template into safety-critical or rule-rich domains. Within ML-system testing, differential-testing approaches such as DeepXplore [36] use input-transformation invariance as a proxy oracle, complementary to the explicit MR formulation we adopt here.

A second strand uses program-structure features rather than test-case mining. Kanewala et al. [23] predict MRs of scientific software via graph-kernel learning over control-flow and data-dependency graphs, an early demonstration that program structure carries enough signal for automated MR discovery. Nolasco et al.’s MemoRIA [35] infers MRs by deriving an object-protocol abstraction and validating candidates through fuzzing plus SAT-based reduction, then evaluates on 22 Java subjects. Tao et al.’s Mettoc [46] applied MT to compiler testing using equivalence-preservation as the canonical relation. Ying et al. [55] formalise relationships between metamorphic-relation patterns into family trees, and Altamimi et al. [1] provide a recent systematic literature review of MR-automation work that catalogues the structural-vs-empirical-search divide our framework targets. Earlier LLM-assisted attempts include Zhang et al. [56] on ChatGPT-driven MR generation.

Each of these methods advances automation on a different axis, but all treat the MR space as an empirical search space. MR-Scout’s recall is bounded by the relations already latent in existing tests.

GenMorph’s evolutionary search is shaped by the fitness landscape induced by mutation killing. LLM-prompted approaches are sensitive to prompt phrasing when no structural prior constrains the generation. None of these methods can state, before search begins, which MR types lie outside its reach, because none has an algebraic account of the space being searched.

2.4 MetaPattern catalogues and empirical adequacy

MetaPattern catalogues organise the post-hoc inventory of identified MRs into a smaller vocabulary of recurring strategies. Such catalogues typically result from clustering observed MRs, codifying expert intuitions about “kinds of properties testers reason about”, and validating the catalogue’s coverage on a benchmark MR corpus. Reactor-physics test suites have produced taxonomies of conservation, monotonicity, convergence, trajectory, and partial-order patterns [7, 26]; recent taxonomic work formalises relationships between MR patterns into family trees [55]. Adequacy frameworks such as the Pattern–Matrix Coverage Metric assess how thoroughly a given MR set occupies the pattern space.

Such catalogues are useful, but each is, by construction, an empirical artefact. None of the existing catalogues, including the present authors’ own, can answer the three foundational questions of Section 1.

2.5 Convergent diagnosis

Across all four lines of work, two problems recur. *Unbounded MR emergence*: every new domain or new program produces fresh MR formulations. *Poor MR reusability*: identified MRs do not transport readily across programs that share deep structural similarities. Both problems follow from grounding MetaPatterns inductively. NOETHER replaces that inductive grounding with operator-algebraic grounding.

3 Operator-algebraic preliminaries

This section introduces the algebraic apparatus used by NOETHER. We assume basic familiarity with group actions, equivalence classes, and quotient sets, but not with functional analysis. Each construct is defined at the level needed for the framework and later instantiated in reactor physics and equivariant machine learning.

3.1 Programs and program-induced operator algebras

Throughout the paper we treat a program P as a (possibly partial) computable function $P : \mathcal{X} \rightarrow \mathcal{Y}$. The class of programs of interest belongs to a *program family* $\mathcal{F}$ whose members share an underlying mathematical scaffolding.

DEFINITION 1 (PROGRAM-INDUCED OPERATOR ALGEBRA). *Let P belong to a program family $\mathcal{F}$. A program-induced operator algebra of $\mathcal{F}$ is a tuple*

$$\mathcal{A}_{\mathcal{F}} = (\mathcal{O}, \circ, \sim_{\mathcal{F}}),$$

where $\mathcal{O}$ is a set of operators acting on $\mathcal{X}$, $\mathcal{Y}$, or both; $\circ$ is an operator composition; and $\sim_{\mathcal{F}}$ is an equivalence relation declaring two operator expressions equal whenever they agree on every program of $\mathcal{F}$.

3.2 Symmetry groups (building block B1)

DEFINITION 2 (SYMMETRY GROUP OF $\mathcal{A}_P$). *A symmetry group of $\mathcal{A}_P$ is a subgroup $G \leq \mathcal{A}_P$ whose elements act on $\mathcal{X}$ such that, for all $P \in \mathcal{F}$ and all $g \in G$,*

$$P(g \cdot x) = \rho(g) \cdot P(x) \quad \forall x \in \mathcal{X},$$

where $\rho : G \rightarrow \text{End}(\mathcal{Y})$ is a (possibly trivial) representation of G on $\mathcal{Y}$.

3.3 Order operators: monotonicity and linearity (building block B2)

DEFINITION 3 (MONOTONE OPERATOR). P is monotone with respect to θ if $\theta_1 \leq \theta_2 \Rightarrow P(\theta_1) \leq_y P(\theta_2)$ (or the reversed inequality, in which case P is anti-monotone).

DEFINITION 4 (LINEAR OPERATOR). P is linear on $\mathcal{X}_0 \subseteq \mathcal{X}$ when $P(\alpha x_1 + \beta x_2) = \alpha P(x_1) + \beta P(x_2)$ for all $x_1, x_2 \in \mathcal{X}_0$ and scalars α, β .

3.4 Self-adjoint operators (building block B3)

DEFINITION 5 (SELF-ADJOINT OPERATOR). Given an inner product $\langle \cdot, \cdot \rangle$ on $\mathcal{X}$, an operator $L \in \mathcal{O}$ is self-adjoint if $\langle Lx, y \rangle = \langle x, Ly \rangle$ for all x, y in the domain of L .

Self-adjointness encodes a duality between the two arguments of an inner product. Reciprocity theorems in physics, transposed-graph identities in algorithms, and detailed-balance conditions in stochastic processes are all instances. In reactor physics, the adjoint transport formulation is self-adjoint under the appropriate inner product, yielding source-detector reciprocity.

3.5 Time-reversal operators (building block B4)

DEFINITION 6 (TIME-REVERSAL OPERATOR). When $\mathcal{X}$ admits a time coordinate, a time-reversal operator $\mathcal{T}$ acts on inputs by reversing the time variable, $\mathcal{T}(x(t)) = x(-t)$. A program P is time-reversal symmetric on a sub-family of inputs when $P(\mathcal{T}x)$ is determined by $P(x)$ through a fixed bijection on $\mathcal{Y}$.

Time-reversal applies only where the underlying dynamics admit a reversible sub-family. Dissipative dynamics break this symmetry, so the corresponding MetaPattern is empty for those systems.

3.6 Limit operators (building block B5)

DEFINITION 7 (LIMIT OPERATOR). A parametrised limit operator $\mathcal{L}_\theta$ is a family of operators indexed by a parameter θ such that there exists a limit element $\mathcal{L}_*$ with $\mathcal{L}_\theta \rightarrow \mathcal{L}_*$ as $\theta \rightarrow \theta_*$ in an appropriate operator topology.

3.7 Qualitative-dynamics operators (building block B6)

DEFINITION 8 (QUALITATIVE-DYNAMICS OPERATOR). A qualitative-dynamics operator $\mathcal{D}$ is an operator on solution trajectories of an underlying ODE/PDE that extracts qualitative features — extrema, inflection points, monotonic phases, overshoot magnitudes, S-curve transitions, phase-portrait orbits — that are invariant under perturbations preserving the underlying dynamical structure (Sturm-type comparison theorems, dynamical-systems classification).

Some MR-relevant invariants are not point-wise but *shape-wise*: a solution curve may have an overshoot, a single extremum, or a monotone-then-saturating profile. B6 gives these qualitative-dynamics MRs their own algebraic root.

3.8 Method-comparison operators (building block B7)

DEFINITION 9 (METHOD-COMPARISON OPERATOR). A method-comparison operator $\mathcal{E}$ encodes a partial order $\leq_{\mathcal{E}}$ on numerical or algorithmic methods, where $M_1 \leq_{\mathcal{E}} M_2$ asserts that method M_1 produces an approximation no worse than method M_2 in a specified error norm, under specified conditions.

3.9 Decomposition of an operator algebra

Given $\mathcal{A}_P = (\mathcal{O}, \circ, \sim_{\mathcal{F}})$, we decompose $\mathcal{O}$ along the eight building blocks introduced above:

$$\mathcal{D}(\mathcal{A}_P) = (G, O_{\leq}, T^*, \mathcal{T}^*, \mathcal{L}^*, \mathcal{D}^*, \mathcal{E}^*, \mathcal{B}_{\text{rel}}^*),$$

where G collects symmetry subgroups, $O_{\leq}$ monotone and linear operators, T^* self-adjoint operators, $\mathcal{T}^*$ time-reversal operators, $\mathcal{L}^*$ limit operators, $\mathcal{D}^*$ qualitative-dynamics operators, $\mathcal{E}^*$ method-comparison operators, and $\mathcal{B}_{\text{rel}}^*$ relational-equivalence operators.

DEFINITION 10 (RELATIONAL-EQUIVALENCE BLOCK). A relational-equivalence operator is a binary relation $\equiv_{\mathcal{R}}$ on expressions over an idempotent semiring $(\mathcal{S}, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ such that $E \equiv_{\mathcal{R}} E'$ iff E and E' are equal under all valid evaluation contexts of $\mathcal{S}$, generated by a finite set $\mathcal{R}_{\text{rel}} = \{\mathcal{R}_1, \dots, \mathcal{R}_K\}$ of identity-preserving rewriting rules, each $\mathcal{R}_i$ an ordered pair $(\text{lhs}_i, \text{rhs}_i)$ of semiring expressions with the property $\text{eval}(\text{lhs}_i, D) =_{\text{bag}} \text{eval}(\text{rhs}_i, D)$ on every valid input D . The block $\mathcal{B}_{\text{rel}}^*$ collects all such operators. It is empty for program families without idempotent-semiring rewriting structure (Boltzmann reactor physics, equivariant ML) and non-empty for those with it (relational query optimisers, §6.7).

Necessity, sufficiency, and the empirical status of the decomposition. We do not claim that the eight blocks are necessary in an absolute sense, that they exhaust all algebraic structures relevant to software testing, or that they follow from first principles. They are an *empirical curation*: a by-inspection enumeration of mathematical structures that recur across the program families we have studied. The claim is therefore scoped: the eight blocks are currently sufficient for these families, not provably necessary in general. This upstream empirical status is NOETHER's main limitation as a theoretical framework. Induction has not been eliminated from MetaPattern discovery; it has been moved one level up, from “what MetaPatterns recur in observed MR samples?” to “what algebraic structures recur across program families?” Given such a curated decomposition, the downstream derivation of $\mathbf{M}(\mathcal{A}_P)$ becomes mechanical and provably closed in the sense of Theorem 1. We state the eight-block sufficiency as an explicit empirical hypothesis with a documented out-of-scope catalogue.

HYPOTHESIS 1 (EIGHT-BLOCK SUFFICIENCY). Let P be a program belonging to a program family $\mathcal{F}$ whose semantics admit an operator-algebraic representation $\mathcal{A}_P$ over (i) finite or finite-dimensional symmetry groups, (ii) partial orders with monotone or linear operators, (iii) self-adjoint operators with respect to a fixed inner product, (iv) anti-unitary involutions of a time coordinate, (v) limit operations of analytical (mesh-refinement, parameter-perturbation, or asymptotic) type, (vi) qualitative-dynamics attractors of an underlying ODE/PDE, (vii) inter-method comparison partial orders, and (viii) identity-preserving rewriting rule sets on an idempotent semiring. For such families the eight-block decomposition

$$\mathcal{D}(\mathcal{A}_P) = G \dot{\cup} O_{\leq} \dot{\cup} T^* \dot{\cup} \mathcal{T}^* \dot{\cup} \mathcal{L}^* \dot{\cup} \mathcal{D}^* \dot{\cup} \mathcal{E}^* \dot{\cup} \mathcal{B}_{\text{rel}}^*$$

is sufficient: every operator in $\mathcal{A}_P$ relevant to MR derivation is assigned to at least one block. Hypothesis 1 is an empirical hypothesis open to refutation.

REMARK 1 (BLOCK SUFFICIENCY VS. Translate SUFFICIENCY). Hypothesis 1 asserts that the eight blocks suffice to assign every operator in $\mathcal{A}_P$ relevant to MR derivation. It does not assert that every MR formulable over $\mathcal{A}_P$'s operators is reachable through the framework's Translate operator from a single block invariant. Section 6.8 exhibits, on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, two MRs whose constituent operators are individually assigned to blocks of $\mathcal{D}(\mathcal{A}_{\text{PWR}})$ (rod-insertion semigroup; boron and moderator-temperature scaling), yet whose MR content is not in $\text{MR}(\mathcal{A}_{\text{PWR}})$ in the sense

of Definition 13. The block decomposition is therefore a necessary but not sufficient input to MetaPattern construction: Translate’s expressive form (single-block invariants, first-order π -template, single partial-order direction, operating on $P(x)$ tuples rather than on operator-spectrum quantities) is a second, independent constraint on the framework’s reach. This distinction between “block sufficiency” (Hypothesis 1) and “Translate sufficiency” (open) is made explicit in §6.8.

REMARK 2 (KNOWN AND CONJECTURED OUT-OF-SCOPE PROGRAM FAMILIES). *The hypothesis is open to refutation. Six families are known or conjectured to fall outside its image and to require an additional block:*

- (1) Symplectic systems. Hamiltonian dynamics whose volume-preserving structure is captured by a symplectic 2-form rather than self-adjointness (N -body simulators, celestial-mechanics integrators).
- (2) Sheaf-theoretic / categorical constructions. Programs whose semantics are captured by functors and natural transformations between categories (certified compilers, type-driven program transformations); MRs of the form “commutativity of a square in the underlying category” do not reduce to single-block invariants under the present Translate.
- (3) Probabilistic / martingale invariants. MRs of the form $\mathbb{E}[f(\mathbf{x}_t) \mid \mathcal{F}_s] = f(\mathbf{x}_s)$ on a stopping-time domain” arise in MCMC samplers and stochastic gradient analyses.
- (4) Topological invariants. MRs of the form “ f ’s level sets have a fixed Betti-number signature” (topological data analysis, shape classifiers) require homological structure absent from the present blocks.
- (5) Label-consistency for supervised learning. MRs that constrain a model’s predictions against ground-truth labels (e.g. wrong-sign-loss detection on classifiers); the absence of a label-consistency operator is one reason the §6.6 case study reports zero detection on category-(i) wrong-sign mutations.
- (6) Empirical parameter-distribution divergence. A probability-distribution divergence operator on parameter-space measures, motivated by the §6.6.1 pilot’s category-v-02/04/05 (activation change, bias removal, weight re-init) which the eight blocks do not detect. An MR of the form $D_{\text{KL}}(p_\theta \| p_{\theta+\Delta\theta}) \leq \tau$ for $\|\Delta\theta\| \leq \epsilon$ would lie in such a block. Empirical witness: §6.6.1’s cat-v-02/04/05 mutations are not detected by any MR set including Set N; this is that pilot’s principal negative finding for Hypothesis 1.

Each family signals a candidate ninth block. The framework’s intended response is to admit the additional block when the program family demands it, rather than to extend $\mathcal{B}_{\text{rel}}^*$ ’s scope or to absorb the case as “out of scope for MR testing” in some over-broad sense. Concrete examples of MRs that lie outside Theorem 1’s scope are catalogued in Appendix D; one candidate ninth block (metric-stability M_{lip}) has an explicit Translate-template design (Remark 3 below).

REMARK 3 (METRIC-STABILITY CANDIDATE NINTH BLOCK, M_{lip}). *Among the candidate ninth blocks of Remark 2, the metric-stability block has the most concrete Translate-template proposal. Define $M_{\text{lip}} = (\mathcal{X}, d_{\mathcal{X}}) \rightarrow (\mathcal{Y}, d_{\mathcal{Y}})$ as the family of K -Lipschitz maps between two metric spaces; the induced MetaPattern m_{lip} is the equivalence class of pointwise stability inequalities $d_{\mathcal{Y}}(P(x'), P(x)) \leq K \cdot d_{\mathcal{X}}(x', x)$. The corresponding Translate template constructs a follow-up by metric perturbation $x' = x + \epsilon u$ ($\|u\| = 1, |\epsilon| < \delta$) and predicates a K -Lipschitz output bound. Canonical-block ordering would place M_{lip} after $\mathcal{B}_{\text{rel}}^*$ since metric structure is independent of the eight existing block invariants (no group action on the perturbation set, no partial order on inputs, no self-adjoint operator, no parametric refinement family, no idempotent-semiring rewriting). Theorem 1’s closure proof transfers without modification because m_{lip} is single-block algebraically derived. We do not commit to M_{lip} as part of the canonical decomposition in this paper; we record it in Appendix D as the most concrete sub-instance*

of Remark 2 item (iv) (topological invariants) that admits an immediate Translate template, and as the empirical orphan in the audit of §5.3.

4 The NOETHER framework

4.1 Algebra-induced metamorphic relations

Before defining algebra-induced MRs we make explicit the Translate operator that converts a single block invariant into an executable MR. Without this definition, the closure result of Theorem 1 would quantify over an undefined construct.

DEFINITION 11 (BLOCK INVARIANT). Let $s \in \mathcal{D}(\mathcal{A}_P)$ be one of the eight blocks. A block invariant of $\mathcal{A}_P$ under s is a pair $\iota = (\Phi, \pi)$ where Φ is a finite set of operators drawn from s , and π is a relation $\pi \subseteq (\mathcal{X} \times \mathcal{Y})^k$ for some arity $k \geq 1$ such that for every $P \in \mathcal{F}$ and every choice of operators $\phi_1, \dots, \phi_n \in \Phi$, the tuple $((x_i, P(x_i)))_{i=1}^k$ obtained by applying $\phi_1, \dots, \phi_n$ to a base input $x \in \mathcal{X}$ in the canonical order specified by s satisfies π . We write $\mathcal{I}_s$ for the set of block invariants under s and $\sim_s$ for the equivalence relation $\iota \sim_s \iota'$ iff $\Phi = \Phi'$ and π, π' define the same constraint up to relabelling of input/output coordinates.

The relation $\sim_s$ is reflexive, symmetric, and transitive by construction (set equality and constraint equality are equivalence relations); hence $\sim_s$ is a genuine equivalence relation on $\mathcal{I}_s$, justifying the quotient in Step 3 of CONSTRUCT-MP below.

DEFINITION 12 (Translate). The Translate operator is the function

$$\text{Translate} : \mathcal{I}_s \times \{s\} \longrightarrow \text{MR}(P)$$

that maps a block invariant $\iota = (\Phi, \pi)$ under block s to the metamorphic relation

$$\rho_{\iota, s} : \quad \forall x \in \mathcal{X}, \forall \phi_1, \dots, \phi_n \in \Phi : \quad \pi((x_i, P(x_i)))_{i=1}^k \text{ holds,}$$

where the input tuple $(x_i)_{i=1}^k$ is generated by applying the ϕ_j in the canonical order specified by s (the convention is fixed per block: e.g., for $s = G$, $x_i = g_i \cdot x_0$ with g_i enumerated by group orbit; for $s = T^*$, the tuple ranges over the two arguments of the inner product). Translate is well-defined because (i) the per-block canonical order resolves any ambiguity in tuple generation, and (ii) $\sim_s$ -equivalent invariants are mapped to MRs that are $\sim$ -equivalent as logical statements over P . Per-block instantiations of Translate are tabulated in Appendix D.

DEFINITION 13 (ALGEBRA-INDUCED MR). Let $\mathcal{A}_P$ be a program-induced operator algebra and let ρ be an MR over a program P . We say ρ is induced by $\mathcal{A}_P$ — written $\rho \in \text{MR}(\mathcal{A}_P)$ — when there exist (i) a block $s \in \mathcal{D}(\mathcal{A}_P)$, (ii) a block invariant $\iota \in \mathcal{I}_s$, and (iii) the equality $\rho = \text{Translate}(\iota, s)$ holds in the sense of Definition 12. Properties of P that constrain executions but cannot be obtained as $\text{Translate}(\iota, s)$ for any single block s and any invariant $\iota \in \mathcal{I}_s$ are not algebra-induced in this sense; concrete examples are catalogued in Appendix D.

4.2 Construction of the MetaPattern set

We present the deductive procedure CONSTRUCT-MP that maps an algebra-decomposition to a MetaPattern set.

Step 1 — Invariant extraction. For each block s in $\mathcal{D}(\mathcal{A}_P)$, compute the set of invariants $\mathcal{I}_s$ of $\mathcal{A}_P$ under s .

Step 2 — MR derivation. For each invariant $\iota \in \mathcal{I}_s$, derive the MR family $\mathcal{R}(\iota) = \{\rho \in \text{MR}(\mathcal{A}_P) \mid \rho = \text{Translate}(\iota', s), \iota' \sim_s \iota\}$.

Step 3 — Quotient. Form the MetaPattern $m_s = \mathcal{R}(\iota) / \sim_s$.

Step 4 — Aggregation. Return $\text{IM}(\mathcal{A}_P) = \{m_s : s \in \mathcal{D}(\mathcal{A}_P)\}$.

4.3 Algebraic closure under Translate, the canonical-block ordering, and out-of-scope MRs

DEFINITION 14 (CANONICAL-BLOCK ORDERING). *We adopt the strict total order on blocks*

$$G > O_{\leq} > T^* > \mathcal{T}^* > \mathcal{L}^* > \mathcal{D}^* > \mathcal{E}^* > \mathcal{B}_{\text{rel}}^*.$$

An MR derivable through multiple blocks is assigned to the highest-priority block in this order. The placement of $\mathcal{B}_{\text{rel}}^*$ at the bottom reflects its semiring-rewriting nature, which sits algebraically downstream of the seven physical-mathematical blocks: rewriting equivalences typically depend on the program family’s input-perturbation, order, and method-comparison structure rather than the converse.

THEOREM 1 (ALGEBRAIC CLOSURE UNDER Translate). *Let $\mathcal{A}_P$ be a program-induced operator algebra with decomposition $\mathcal{D}(\mathcal{A}_P)$ as in Section 3.9, and let $\mathbb{M}(\mathcal{A}_P) = \text{CONSTRUCT-MP}(\mathcal{D}(\mathcal{A}_P))$. Then for every $\rho \in \text{MR}(\mathcal{A}_P)$ in the sense of Definition 13, there exists a unique $m \in \mathbb{M}(\mathcal{A}_P)$ such that $\rho \in m$, where uniqueness is determined under the canonical-block ordering.*

REMARK 4 (SCOPE OF THEOREM 1). *Theorem 1 is a closure statement, not a completeness claim over arbitrary properties one might assert about P ’s executions. It quantifies over $\text{MR}(\mathcal{A}_P)$ as defined by Definition 13 — the algebra-induced MRs reachable through the Translate operator from a single block invariant. Three concrete classes of MRs lie outside this scope and are not covered by the theorem; they are documented in Appendix D:*

- (1) Probabilistic MRs without operator-algebraic representation — e.g. MRs that constrain output distributions (such as “the entropy of $f(\mathbf{x})$ should not decrease under input augmentation A ”) when the augmentation is not expressible as an operator in $\mathcal{O}$.
- (2) Input-distribution MRs not expressible over $\mathcal{A}_P$ operators — e.g. adversarial-perturbation MRs “ $\|\delta\|_p \leq \epsilon \Rightarrow f(\mathbf{x} + \delta) = f(\mathbf{x})$ ” whose perturbation set is not a group action on $\mathcal{X}$.
- (3) Compositional MRs spanning multiple blocks under non-Translate derivations — MRs requiring simultaneous use of two block invariants in a way that is not the canonical-block-ordering reduction of Definition 14.

The strictly stronger statement that every MR formulable as a property over $\mathcal{A}_P$ ’s operators (without restricting to Translate-reachable derivations) is contained in some $m \in \mathbb{M}(\mathcal{A}_P)$ is identified as Theorem 1’ in Appendix D. Section 6.8 and Appendix D establish that this stronger statement is false on the PWR core diffusion algebra $\mathcal{A}_{\text{PWR}}$, by exhibiting two concrete counterexamples (ρ_{nonadd} , $\rho_{\text{MTC-bor}}$) whose obstructions identify five structurally independent extensions of Translate’s signature. The combined open problem (whether such an extended Translate preserves Theorem 1’s closure and Theorem 2’s polynomial-time decidability) is the principal open question for follow-up work.

A sceptical reading might object that the by-construction status of Theorem 1 makes it near-tautological. We accept this characterisation for the literal content of the theorem. The substantive value lies elsewhere: Theorem 1 converts an empirical-adequacy claim (“our pattern grid covers $X\%$ of observed MRs”) into a structural-adequacy claim (“our pattern grid is exhaustive of the algebra-induced MR space, modulo the explicit out-of-scope classes of Remark 4”). Empirical adequacy frameworks built on inductive pattern grids do not guarantee algebraic closure. NOETHER guarantees algebraic closure, and the obligation imposed on any user of the framework is then to verify that no Translate-reachable MR is dropped by CONSTRUCT-MP, a verification that is mechanical once the algebraic input has been fixed.

The theorem still imposes a checkable obligation on the framework. If an MR is induced by an invariant in $\mathcal{D}(\mathcal{A}_P)$ but cannot be assigned by CONSTRUCT-MP, then at least one component, invariant extraction, Translate, structural equivalence, or the canonical ordering, is underspecified.

Thus the result is not an empirical coverage claim; it is a well-formedness and closure guarantee for the downstream construction once the algebraic input has been fixed.

REMARK 5 (SCOPE OF THEOREM 1 FOR THE $\mathcal{B}_{\text{rel}}^*$ BLOCK). *The closure result extends to $\mathcal{B}_{\text{rel}}^*$ provided the rewriting rule set $\mathcal{R}_{\text{rel}}$ is finite. For the relational-algebra fragment of TPC-H-class queries the rule set is finite by the standard heuristic-rewrite normal forms (selection-pushdown, projection-pushdown, join-reordering, distinct-elimination) [50, 60], and Theorem 1 therefore covers algebra-induced MRs of this fragment. Query languages with unbounded recursive rewriting (e.g. Datalog with non-terminating rule sets) lie outside this scope and are catalogued as out-of-scope under Remark 2.*

4.4 Decidability and complexity

THEOREM 2 (DECIDABILITY). *Suppose $\mathcal{A}_P$ admits a finite generating set $\text{gen}(\mathcal{A}_P)$ of cardinality n , with each generator's invariant computation taking time t_i . Then $\text{M}(\mathcal{A}_P)$ is computable in time $O(n \cdot \max_i t_i \cdot \log n)$.*

REMARK 6 (SCOPE OF THEOREM 2 FOR THE $\mathcal{B}_{\text{rel}}^*$ BLOCK). *The polynomial-time bound holds for $\mathcal{B}_{\text{rel}}^*$ when the rewriting rule set $\mathcal{R}_{\text{rel}}$ is finite. For first-order SQL with bag semantics, query equivalence is undecidable in general, but query equivalence under a fixed rule set $\mathcal{R}_{\text{rel}}$ is decidable: practical SMT-based solvers verify equivalence on substantial benchmark fractions in the published literature [50, 60]. Theorem 2 therefore covers the rule-set-bounded fragment of relational algebra; outside this fragment, decidability is governed by the underlying first-order theory and is not the framework's responsibility.*

Table 1. Per-generator cost of invariant extraction in each block. The symmetry block (G) is split by group regime: finite, finite-dimensional Lie, and finitely generated infinite-discrete (with user-supplied truncation K). For Lie groups (e.g. $\text{SO}(3)$, $d_G = 3$) the bound $O(|G|^2)$ does not apply because $|G|$ is uncountable; the relevant bound is $O(d_G^2)$ over the Lie-algebra basis. See §4.4 for the truncation discussion.

Block	Regime	t_i for one generator
G (symmetry)	finite group	$O(G ^2)$ (group-orbit fixed-point)
	finite-dim. Lie group	$O(d_G^2)$, $d_G = \dim_{\mathbb{R}} \mathfrak{g}$
	infinite discrete (truncated)	$O(K^2)$ at truncation $ G \leq K$
$O_{\leq}$ (order)	—	$O(n^2)$ (poset comparison)
T^* (self-adjoint)	—	$O(d)$ (inner-product symmetry check)
$\mathcal{T}^*$ (time-reversal)	—	$O(1)$
$\mathcal{L}^*$ (limit)	—	$O(\log_{\epsilon} \frac{1}{\epsilon})$
$\mathcal{D}^*$ (qualitative-dynamics)	—	$O(d)$
$\mathcal{E}^*$ (method-comparison)	—	$O(K^2)$ (over K methods)

Per-block invariant-extraction cost.

On infinite groups. The first row of Table 1, $O(|G|^2)$, is the natural bound for finite symmetry groups and is the regime under which Theorem 2 was originally stated. For a Lie group G such as $\text{SO}(3)$, $|G|$ is uncountable and the bound is replaced by $O(d_G^2)$, where $d_G = \dim_{\mathbb{R}} \mathfrak{g}$ is the real dimension of G 's Lie algebra. In the equivariant-ML instantiation of Section 6, $d_{\text{SO}(3)} = 3$; the symmetry-block invariant is computed once over a basis of three infinitesimal generators, and orbit closure is obtained through finite-dimensional linear algebra. For finitely generated infinite discrete groups (e.g. $\mathbb{Z}$, the integer-translation group of an unbounded grid solver), CONSTRUCT-MP requires a separate truncation parameter K at which orbits are enumerated; the cost is $O(K^2)$.

per generator, and the framework's user is responsible for justifying that the truncated set captures the program family's structurally relevant invariants. We do not claim Theorem 1's closure result is preserved across truncation: closure is over $\text{MR}(\mathcal{A}_P)$ defined relative to the truncated group, and the user must inherit responsibility for whether this is the intended algebra.

For the Boltzmann instantiation the relevant generators are finite ($n \leq 14$, geometric quarter-rotations and energy-group permutations); for the equivariant-ML instantiation the relevant generators are finite-dimensional Lie ($d_{\text{SO}(3)} = 3$, $|\mathfrak{S}_n| = n!$ truncated to a single generator class, $n \leq 10$ training-size, depth, and dimension limits). The asymptotic bound is therefore not the binding constraint in either instantiation.

4.5 The principal limitation

NOETHER replaces inductive grounding with algebraic grounding *downstream of* $\mathcal{A}_P$. Upstream, the distillation of $\mathcal{A}_P$ from a program family remains a human task. This limitation is central to the framework's scope: NOETHER does not automate domain modelling, but it turns the step after domain modelling into a well-defined construction. A second limitation, made explicit in §6.8 and Appendix D, is that even when $\mathcal{A}_P$ is fully specified, Translate's present signature (single-block, first-order π -template, single partial-order direction, operating on $P(x)$ tuples rather than on operator-spectrum quantities) systematically excludes a class of MRs that engineering practice treats as standard: non-additivity of operator-composition functionals and higher-order mixed parametric dependences. The principal limitation is therefore not only the upstream distillation of $\mathcal{A}_P$ but also the present signature of Translate.

Boundary of contribution (Section 4 restatement)

The downstream layer is mechanical: Theorem 1 (algebraic closure under Translate) and Theorem 2 (polynomial-time decidability) operate on a given block decomposition. The upstream layer remains empirical: the eight-block decomposition is Hypothesis 1, an open empirical hypothesis with six documented out-of-scope program-family classes (Remark 2). Theorem 1' (absolute completeness) is open. Sections 5, 6, and 6.7 instantiate NOETHER on three structurally distinct program families.

5 Boltzmann instantiation: from transport to diffusion to burnup

This section instantiates NOETHER on the Boltzmann transport equation and traces the construction through neutron transport, diffusion, and burnup.

5.1 The Boltzmann program family and its operator algebra

The Boltzmann transport equation governs the distribution of neutral particles. In its time-independent eigenvalue form for fission systems:

$$\begin{aligned} \hat{\Omega} \cdot \nabla \psi(\vec{r}, \hat{\Omega}, E) + \Sigma_t(\vec{r}, E) \psi \\ = \iint \Sigma_s(\vec{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega}) \psi dE' d\hat{\Omega}' + \frac{1}{k} \chi(E) \int v \Sigma_f(\vec{r}, E') \psi dE'. \end{aligned} \quad (1)$$

The program-induced operator algebra $\mathcal{A}_{\text{Boltz}}$ contains $\{G_{\text{geom}}, \mathfrak{R}_E, \mathcal{L}_\Sigma, \mathcal{L}_v, \mathcal{L}^*, \mathcal{T}, \mathcal{L}_h, \mathcal{D}_{\text{Bate}}, \mathcal{E}_{\text{cmp}}\}$. Decomposed along the eight blocks of Section 3.9, $\mathcal{A}_{\text{Boltz}}$ yields entries in seven blocks (with $\mathcal{B}_{\text{rel}}^*$ empty under the absence of idempotent-semiring rewriting structure). Every operator listed has an unambiguous status in the established theory of neutral-particle transport [7, 26].

5.2 Running CONSTRUCT-MP on $\mathcal{A}_{\text{Boltz}}$

Steps 1–4 of CONSTRUCT-MP yield seven MetaPatterns: m_{inv} (invariance/equivariance), m_{mono} (parameter-monotonicity), m_{adj} (self-adjoint duality / adjoint reciprocity), m_{rev} (time-reversal compatibility), m_{conv} (discretisation convergence), m_{dyn} (qualitative-dynamics shape invariants), and m_{cmp} (method-comparison error-bound partial orders).

5.3 Relationship to the prior inductive catalogue: refinement plus prediction

A reactor-physics MetaPattern catalogue distilled from the standard PWR-physics literature [7, 26] identifies five MetaPatterns (P1–P5) inductively. The relationship with $\mathcal{M}(\mathcal{A}_{\text{Boltz}})$ is more structured than a simple bijection. NOETHER reproduces three of the prior patterns, refines two on a sounder algebraic basis, and predicts two structurally distinct classes that the inductive catalogue did not isolate:

Table 2. Relationship between prior inductive catalogue and NOETHER deductive output.

Prior catalogue	NOETHER	Relationship
P1 conserva- tion/invariance	m_{inv}	Reproduced — G -symmetry invariants project to the same MRs.
P2 monotonicity	m_{mono}	Reproduced.
P3 convergence	m_{conv}	Reproduced.
P4 trajectory	m_{dyn}	Refined — inductive P4 conflated qualitative-dynamics with time-reversal. NOETHER places trajectory phenomena in m_{dyn} (Sturm-type comparison theorems) and exposes the conflation.
P5 partial- order/bounding	m_{cmp}	Refined — inductive P5 grouped method-accuracy partial orders with adjoint reciprocity. NOETHER places the former in m_{cmp} (approximation-theory error bounds) and exposes the distinction.
(none)	m_{adj}	Predicted — adjoint-reciprocity MRs derived as a structurally distinct equivalence class.
(none)	m_{rev}	Predicted — collisionless-trajectory-reversal MRs derived for the corresponding sub-formulations.

This is not the structure of a re-coding. NOETHER’s output structurally *refines* two prior patterns and *predicts* two additional patterns the inductive method missed. The gain is canonical placement: two previously conflated inductive labels are separated by algebraic source, and two textbook phenomena become mandatory MetaPattern classes once their blocks appear in $\mathcal{D}(\mathcal{A}_{\text{Boltz}})$.

A note on prediction (and an interpretive caveat). We do not claim that m_{adj} (adjoint reciprocity) and m_{rev} (collisionless time-reversal compatibility) are de novo physical discoveries. Adjoint-flux reciprocity is standard textbook material in transport theory [7, 26], and time-reversal MRs in collisionless transport have been understood in physics for decades. The MT-community-level absence of these MetaPatterns from the prior inductive catalogue reflects which phenomena happened to be canonically encoded in the 84-MR corpus, not the absence of the underlying physics from the literature.

We must also acknowledge an interpretive caveat about what this “prediction” is and is not. The blocks T^* (self-adjoint) and $\mathcal{T}^*$ (time-reversal) of Section 3.9 were themselves curated by inspection of program families that include reactor physics; the operators $\mathcal{T}_{\text{geom}}$ and $\mathcal{T}$ in $\mathcal{A}_{\text{Boltz}}$ were the leading examples motivating the inclusion of those blocks. There is therefore a circularity in the strong reading of “prediction”: T^* and $\mathcal{T}^*$ were partly induced from reactor-physics

structures, and m_{adj} and m_{rev} are then derived from those blocks. The framework does not discover these MetaPatterns *de novo*. What it does is closer to a uniform *re-projection*: given the eight-block decomposition (whatever its empirical provenance), CONSTRUCT-MP places adjoint-reciprocity and time-reversal phenomena into structurally distinct equivalence classes that the inductive 84-MR catalogue had not isolated as such, even though their underlying physics had been individually known to domain experts. The substantive contribution is the *re-classification under a uniform algebraic structure*, which gives a future testing toolchain a principled reason to enumerate m_{adj} and m_{rev} alongside m_{inv} , m_{mono} , m_{conv} , m_{dyn} , and m_{cmp} rather than treating them as ad-hoc additions. Where NOETHER's predictive power is genuine and non-circular is in the deflationary direction (Section 8.3): the framework can also reveal that an existing inductive catalogue *over-counts* structurally distinct patterns. Both directions, re-classification and de-duplication, are systematisation, not discovery.

Element-wise correspondence. Table 3 traces 12 representative MRs distilled from the standard reactor-physics literature [7, 26] to their NOETHER placement. Appendix B documents each row's source equation, Translate template, and block assignment with explicit textbook citations. The 12 MRs were selected by the following protocol so that the table is structurally rather than numerically representative: (i) every block $s \in \mathcal{D}(\mathcal{A}_{\text{Boltz}})$ that is non-empty is represented at least once; (ii) within each block, the most frequent canonical-form MR in the literature is selected first; (iii) where a block contains MRs from two distinct sub-categories (e.g. geometric vs. energy-group symmetry within G), each sub-category is given at least one representative; (iv) the two predicted MetaPatterns m_{adj} and m_{rev} are listed at the bottom in italics to indicate that no MR was previously catalogued in this form; the block placement is derived. A larger corpus underlying the protocol is provided as supplementary material S2.

An independent audit on the 18-MR engineering catalogue. The 12-row table is structurally curated. As a coarser breadth check, an independent 18-MR engineering catalogue distilled from production reactor-physics codes (orthogonal to the 84-MR inductive corpus underlying Section 5.3) was labelled by three independent large language models against the seven canonical MetaPatterns plus the relational extension of Section 3.9, with a fourth label *orphan* reserved for MRs that fit none of the eight classes; the labelling protocol, raw labels, and majority-vote tabulation are released in supplementary material S2 (18mr_audit/). Inter-rater agreement on the four-way classification is almost-perfect (Fleiss' $\kappa = 0.857$, $n = 18$ items, $r = 3$ raters, $c = 4$ categories), and 17 of the 18 MRs are placed by majority vote into one of the seven canonical MetaPatterns or m_{rel} (subsumption 94.4%, Wilson 95% CI [74.2%, 99.0%] on the binary subsumption proportion). The single orphan is the Lipschitz / metric-stability MR, which we treat structurally in Appendix C.5.2 as a candidate ninth block rather than as a counter-example to Theorem 1's scope. The independence of the three labellers is bounded by the LLM-shared-training-data caveat (the three models share substantial pre-training corpora, so κ should be read as agreement among similarly-trained but non-coordinated raters rather than among rigorously independent oracles); we report the audit as external corroboration of breadth rather than as an independent verification of correctness.

5.4 A Noether-style derivation of m_{adj}

The framework's name is methodological. Noether's first theorem replaces an empirically curated catalogue of conservation laws with a derivation from the symmetry structure of an action functional [34]. We do not invoke Noether's theorem as a theorem about programs (programs do not, in general, possess an action functional in the variational-calculus sense). What the framework does mirror is the *methodological move*: instead of cataloguing observed invariants, derive them from a structural source. We illustrate the move on m_{adj} .

Table 3. Twelve representative MRs from the prior PWR corpus, with NOETHER placement.

MR ID	Plain-text MR	P#	Block	NOETHER MP
Bur-Phy-01	Step-splitting invariance	P1	G	m_{inv}
Bol-Phy-02	Quarter-symmetry rotation	P1	G	m_{inv}
Bol-Phy-11	$\Sigma_a \uparrow \Rightarrow k_{\text{eff}} \downarrow$	P2	$O_{\leq}$	m_{mono}
Bol-Phy-12	$\nu \Sigma_f \uparrow \Rightarrow k_{\text{eff}} \uparrow$	P2	$O_{\leq}$	m_{mono}
Dif-Alg-01	Diamond-difference h^2 convergence	P3	$\mathcal{L}^*$	m_{conv}
Bur-Alg-01	CRAM order $N \rightarrow \infty$ convergence	P3	$\mathcal{L}^*$	m_{conv}
Bur-Phy-08	Iodine pit qualitative shape	P4	$\mathcal{D}^*$	m_{dyn}
Cpl-App-06	Gd self-shielding S-curve	P4	$\mathcal{D}^*$	m_{dyn}
Bur-Alg-04	CRAM no-worse-than TTA	P5	$\mathcal{E}^*$	m_{cmp}
Bol-Alg-04	P_0 over-estimates k in H-systems	P5	$\mathcal{E}^*$	m_{cmp}
(predicted)	Adjoint reciprocity	—	T^*	m_{adj}
(predicted)	Collisionless reversibility	—	$\mathcal{T}^*$	m_{rev}

Consider the bilinear form on solution-adjoint pairs

$$\mathcal{F}[\phi, \phi^\dagger] = \langle \phi^\dagger, B\phi \rangle - \langle \phi, B^\dagger \phi^\dagger \rangle, \quad (2)$$

where B is the Boltzmann operator and $B^\dagger$ its formal adjoint under the inner product $\langle f, g \rangle = \int f g d\Omega dE d\mathbf{r}$. The bilinear form $\mathcal{F}$ is identically zero on solutions of the forward and adjoint equations: $B\phi = S$, $B^\dagger \phi^\dagger = S^\dagger$ with appropriate boundary conditions imply $\mathcal{F}[\phi, \phi^\dagger] = \langle \phi^\dagger, S \rangle - \langle \phi, S^\dagger \rangle$, and the standard reciprocity identity makes the right-hand side zero. The vanishing of $\mathcal{F}$ is a *conserved current* in the Noether sense: it is annihilated by the dual symmetry $\phi \leftrightarrow \phi^\dagger$, $B \leftrightarrow B^\dagger$. CONSTRUCT-MP's Step 1 extracts $\mathcal{F} = 0$ as the T^* -block invariant; Translate converts it into the executable MR

$$\rho_{\text{adj}} : |\langle \phi^\dagger, S \rangle - \langle \phi, S^\dagger \rangle| \leq \tau, \quad (3)$$

which the testing harness checks on a forward and an adjoint solver run with externally specified sources $S, S^\dagger$. The MR is therefore a Noether-style consequence of the duality symmetry rather than a manually catalogued reactor-physics property: the symmetry $\phi \leftrightarrow \phi^\dagger$ is the structural source, the bilinear form is the conserved current, and the MR is the conservation law in testable form. The same template applies to m_{rev} with the involution $\mathcal{T} : \phi(\mathbf{r}, \boxminus, E, t) \mapsto \phi(\mathbf{r}, -\boxminus, E, -t)$ as the symmetry. The template's general statement is: given a symmetry of the program family's operator algebra, CONSTRUCT-MP extracts the invariant of the symmetry and Translate converts it into an executable MR.

5.5 Specialisation to neutron transport, diffusion, and burnup

The Boltzmann formulation is the most general; specialised solvers are obtained by truncating or projecting $\mathcal{A}_{\text{Boltz}}$. Neutron transport solvers retain the full angular dependence; the same seven MetaPatterns apply. Neutron diffusion solvers truncate angular dependence to its P_1 approximation: the time-reversal block contracts (diffusion is dissipative), removing m_{rev} . Burnup-coupled solvers introduce the Bateman ODEs; the Bateman operator e^{At} enriches multiple blocks (G semigroup, $O_{\leq}$ linearity, $\mathcal{D}^*$ overshoot/S-curve, $\mathcal{E}^*$ CRAM-vs-TTA bounds).

The MetaPattern set is *compositional* with respect to the underlying algebra: add a block, gain MetaPatterns; remove a block, lose them. The framework supports not only construction but also *prediction* of which patterns a new specialisation will exhibit, before any MR is identified empirically.

5.6 Summary

NOETHER, applied to $\mathcal{A}_{\text{Boltz}}$, deductively produces a seven-MetaPattern set that refines the prior inductive catalogue and predicts two additional MetaPatterns (m_{adj} , m_{rev}). The next section turns to a more demanding case: a domain in which the inductive catalogue does not yet exist.

6 Cross-domain demonstration: equivariant machine learning

6.1 The equivariant-ML program family and its operator algebra

Equivariant neural networks impose by architectural construction that certain symmetries of the input induce predictable transformations of the output [8, 12, 24, 47]. The program family $\mathcal{F}_{\text{equi}}$ comprises classifiers satisfying $f(g \cdot \mathbf{x}) = \rho(g) \cdot f(\mathbf{x})$ for $g \in G = \text{SO}(3) \times \mathfrak{S}_n$. Decomposed along the eight blocks: $G = \{G_{\text{equi}}\}$, $O_{\leq} = \{O_{\leq}^{\text{train}}\}$, $T^* = \{T_{\text{att}}^*\}$, $\mathcal{T}^* = \{\mathcal{T}_{\text{seq}}\}$, $\mathcal{L}^* = \{\mathcal{L}_{\text{train}}, \mathcal{L}_{\text{depth}}, \mathcal{L}_{\text{dim}}\}$, $\mathcal{D}^* = \emptyset$ (for feedforward classifiers), $\mathcal{E}^* = \emptyset$ within a single architecture, $\mathcal{B}_{\text{rel}}^* = \emptyset$ (no idempotent-semiring rewriting on equivariant-classifier outputs).

6.2 Running CONSTRUCT-MP on $\mathcal{A}_{\text{equi}}$

CONSTRUCT-MP returns

$$\mathbb{M}(\mathcal{A}_{\text{equi}}) = \{m_{\text{inv}}^{\text{eq}}, m_{\text{mono}}^{\text{eq}}, m_{\text{adj}}^{\text{eq}}, m_{\text{rev}}^{\text{eq}}, m_{\text{conv}}^{\text{eq}}\}.$$

The labels mirror those of $\mathbb{M}(\mathcal{A}_{\text{Boltz}})$ but the *content* is domain-specific. This section does not claim empirical validation in machine learning. It checks a narrower transfer claim: once $\mathcal{A}_{\text{equi}}$ is specified, the same downstream construction yields domain-specific MR families without using a reactor-physics corpus.

6.3 End-to-end derivation: a concrete MR for SE(3)-equivariant point-cloud classification

To illustrate the framework's generative use, we trace a complete derivation from algebra to executable MR.

Step 1. System under test. Let $f : \mathbb{R}^{n \times 3} \rightarrow \Delta^{C-1}$ be a point-cloud classifier mapping n three-dimensional points to a probability distribution over C classes.

Step 2. Distil $\mathcal{A}_{\text{equi}}$. The relevant blocks are $G = \text{SO}(3) \times \mathfrak{S}_n$ and $\mathcal{L}^*$ for training-size and depth limits.

Step 3. Run CONSTRUCT-MP, select an invariant. Within G 's symmetry block, fix attention on the rotation invariant: $f(R \cdot \mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x}$ and $R \in \text{SO}(3)$.

Step 4. Translate the invariant into an executable MR..

$$\rho_{\text{rot}} : \quad \forall R \in \text{SO}(3), \quad \|f(R \cdot \mathbf{x}) - f(\mathbf{x})\|_{\infty} \leq \tau, \quad (4)$$

with $\tau = 10^{-4}$ for fp32 architectures.

Listing 1. Executable MR ρ_{rot} for SE(3)-equivariant classifier.

Step 5. Generate an executable test.

```

1 import numpy as np
2 from scipy.spatial.transform import Rotation
3
4 def test_rotation_invariance(model, point_cloud, num_samples=100, tau=1e-4):
5     """Executable MR rho_rot for SE(3)-equivariant classifier."""
6     p_original = model.predict(point_cloud)
7     failures = []

```